

Functional Specification Document:

Phase 1 (MVP)

Project: Career Counselling Platform for student cohort of Class 9–10

Client: Phoenix Water Club (PWC)

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Date: Jul 20, 2026

1. Project Overview

Phoenix Water Club (PWC) is commissioning a custom, web-based platform to digitize and automate its multi-stage career counseling program. The platform operates on a **B2B model**, servicing educational institutions. Phase 1 focuses on delivering a Minimum Viable Product (MVP) within a targeted ~30-day timeline to facilitate pre-booked institutional orders. The core objective of Phase 1 is to establish a seamless, manual-input-driven digital lifecycle for students, parents, and counselors, setting the foundation for advanced automations (Phase 2 & 3). Architecture - a modular monolith for Phase 1 to support white labelled subscription business models.

Phase 1 scope

- Admin, Counsellor, Student panels (three fixed roles)
- Bulk student upload, credential generation
- Profile form, pre-counselling forms (student + parent link-based)
- Cohort-specific MCQ assessment with PWC-supplied scoring logic
- Auto-generated, live-editable Counsellor Chart
- Session scheduling (combined calendar, same-counsellor mapping, reschedule/cancel)
- Career Library (browse/search + manual add with admin approval)
- Feedback forms (student + parent) with report-gating logic
- Career IKIGAI Report generation (PDF) + simplified parent/institution summaries
- Data purge/retention job

2. User Roles & Access

The system will support three primary user access levels.

Role	Access & Capabilities
Admin (PWC Staff)	Manages student credentials, schedules, report distribution, overall platform monitoring, and career library data.
Counsellor	Views assigned student data, edits the live Counsellor Chart during sessions, and accesses the Career Library.
Student	Accesses forms, assessments, session booking, the Career Library, and the final Career Path Report.
Super Admin	Manages platform-wide settings, onboards new institution (tenant) accounts, and manages career library data.

Note: Parents do not have dedicated login credentials. They will interact with the platform via secure, time-limited, email-specific links.

3. Platform Workflow & Functional Requirements

Tenant Onboarding

- The Super Admin accesses the platform-wide management panel to provision new institution (tenant) accounts.
- This step includes configuring initial settings for the institution and establishing the primary login credentials for the institution's designated Admin.

Student Onboarding & Credentials

- Admin receives a bulk roster of participating students from the client institution via Excel (Name, Class, Contact, Email).
- Admin generates unique login IDs (e.g., CB1, CB2) and default passwords for each

student.

- Admin exports these credentials to be shared with the institution for distribution.

Profile Completion & Security

- Students log into the platform using the provided credentials.
- The system forces a mandatory password change upon the first successful login.
- The student completes a "Profiling Form" capturing basic personal and parental details.
- The student designates one primary email ID and one primary mobile number for each parent.

Pre-Counselling Data Collection

- The student accesses and completes an 80% MCQ / 20% Open-ended Pre-Counselling form.
- The system dispatches a secure, non-editable, time-expiring link to the parent's primary email.
- The parent completes their independent Pre-Counselling form via the link.
- The system restricts progression to Step 6 until both student and parent forms are submitted.

Psychometric Assessment

- The student unlocks a 25–30 minute, MCQ-based assessment.
- The system processes the responses using PWC's proprietary scoring logic to output personality traits and recommended career streams.

The Counsellor Chart (Core Workspace)

- The system automatically generates a unified "Counsellor Chart" upon assessment completion.
- The chart displays student pre-counselling responses, parent pre-counselling responses, and assessment results side-by-side.
- This document is strictly visible to the assigned counselor and acts as the raw data source for all future reports.

Session Scheduling

- Counselors submit their monthly availability (between 8:00 AM and 8:00 PM, Monday–Saturday) to the Admin.

- The system compiles a master calendar of available slots.
- The student selects an open slot for Session 1 without seeing the counselor's identity.
- The system assigns the counselor linked to that slot.
- The system filters all future availability to only show slots belonging to that *same* counselor for the booking of Session 2.
- Built-in functionality for cancellation and rescheduling by either party with backend notifications.

Career Library

- A searchable, browsable database containing 1,300+ career roles mapped by domain, qualification, entrance exams, and average salaries.
- Accessible to both students (for exploration) and counselors (for mapping to the chart).
- Counselors can manually research unlisted careers on external search engines and paste the data into the Counsellor Chart using a prescribed format.
- Any externally added careers trigger a notification to both the Admin and Super Admin for potential ratification and permanent inclusion in the central library.

Live Sessions (1 & 2)

- Sessions occur over third-party video conferencing (e.g., Zoom links manually shared by Admin).
- The counselor actively edits the Counsellor Chart during the discussion, modifying identified strengths/hobbies and documenting career recommendations (narrowing down from 6 to 2 careers).
- The system maps specific fields from the updated Counsellor Chart to a dynamic "Draft Report" visible to the student.
- Session 2 utilizes the updated chart to finalize the student's career path.

Feedback & Report Generation

- The student completes a mandatory Feedback Form via their dashboard.
- The parent completes a Feedback Form via a securely emailed link.
- Upon submission of both forms, the system generates the final reports.
- The student unlocks and downloads the "Student Career Path Report."
- The system generates separate, simplified summary templates for the Parent (sent via email) and the Institution.

4. System Architecture

Recommended Architecture: A modular monolith for Phase 1—designed for rapid deployment within 30 days while ensuring strong data consistency and maintaining clean module boundaries to facilitate future decomposition into microservices.

High-Level Flow

Student / Parent / Counsellor / Admin → REST API Layer → Application Server (modular monolith) → PostgreSQL Database → Report Generation & Storage

Architectural Layers

- **Authentication & Role Management:** JWT-based; supports Admin, Counsellor, and Student roles; parent access via secure, expiring tokens.
- **Workflow / Status:** Enforces the Step 1–12 lifecycle and stage-gating business rules (e.g., assessment locks).
- **Assessment & Scoring:** Processes MCQ assessments using PWC-supplied logic.
- **Session Scheduling & Calendar:** Manages counselor availability and booking logic.
- **Notification Layer:** Transactional email service for link delivery and reminders (WhatsApp/SMS deferred to Phase 2).
- **Report Generation & Document Management:** Handles PDF/Excel exports and document storage.
- **Analytics & Dashboard:** Centralized monitoring of platform health and user activity.
- **Security & Audit Logging:** Ensures compliance, data integrity, and traceability.

Integrations (Phase 1)

- **Cloud Storage:** Object storage for reports, assessment files, and session documents.
- **Transactional Email:** Service for secure links, OTPs, and reminders.
- **Video Conferencing:** Supported via manually shared links (Zoom or equivalent); no direct API integration in Phase 1.

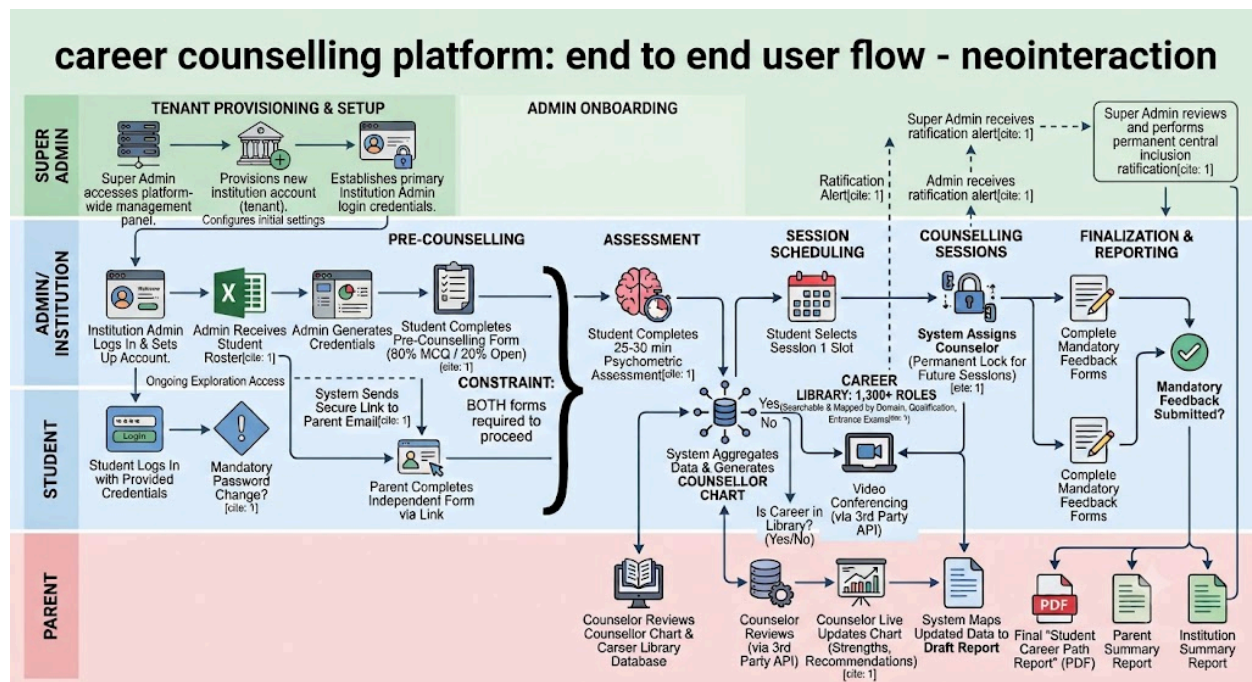
White-Labeling & Subscription Model

- **Platform:** The architecture supports a white-labelled, subscription-based model for institutions and independent counsellors, with no PWC/kREATE branding visible to clients.
- **Admin Management:** The Admin Panel includes an Institution/Subscription management view to track active institutions, renewal dates, seats, and career-library-sync status.
- **Career Library Control:** Centralized ownership and control of the Career Library remain with PWC. Subscription access ensures clients always have the latest library updates.

- **Sync Mechanism:** Scheduled API-based synchronization refreshes each white-labelled instance from the PWC master version, maintaining up-to-date data without manual re-uploads.

Admin Dashboards

- Student statistics (total, active, pending registration, assessments completed, sessions completed, feedback pending, reports generated) match a manual count on a test dataset
- Counsellor statistics (total, available, sessions completed, upcoming, pending reports) match manual counts
- Institution-wise student counts are correct when multiple institutions are loaded simultaneously



This diagram is to represent the overall flow across various users & functional modules. There may be feature overlap between later phases which need to be considered while reviewing.

8. Client Dependencies & Milestones

To ensure the timely delivery of the Phase 1 MVP within the 30-day timeline, Phoenix Water Club (PWC) is expected to provide the following materials and data:

- **Branding and style references:** App UI will be styled using any specified visual guidelines or using any sample references.
- **Form Questionnaires:** The exact questions and formats for the Profiling Form, Student

Pre-Counselling Form, Parent Pre-Counselling Form, and Feedback Forms.

- **Assessment Logic & Content:** The complete question bank for the psychometric assessment, along with the proprietary scoring logic and rules for career mapping.
- **Career Library Database:** The comprehensive Excel sheet containing the 1,300+ career roles mapped across domains, qualifications, entrance exams, and salaries.
- **Report Templates:** The pre-defined output templates/formats for the Counsellor Chart, Student Career Path Report, Parent Report, and Institutional Report.
- **Initial Setup Data:** The first set of student lists (Excel format) and counselor availability schedules to facilitate initial platform testing and onboarding.
- **Domain & Hosting Details:** Access to the domain registrar (e.g., phoenixwaterclub.com) and preferred hosting environment details, if already procured.

Milestones

Milestone	Working Days	Tasks
1	0	Contract sign/Advance
2	12	Frontend UI Screens and flow
3	13	Backend API generation / Integration
4	3	Testing and Going Live

*Working days are considered from mon-friday. Any holidays also will be excluded.
Any delay with milestone completion will be communicated to the client team prior with the backup plans.*

9. Budget & Other Costs

	Items	Approx INR
1	Server Hosting Costs (India/Asia region)	4.5k-9k/month
2	Third party plugins(on a need basis)	TBD

Above pricing is only indicative for budgeting purposes.

5. Technology Stack

Based on the requirements for scalability and rapid MVP deployment, the following technology stack is proposed for Phase 1:

- **Frontend Stack:** React 18, Vite, TypeScript, Zustand, and styled-components.
- **Backend Stack:** Node.js, Express, TypeScript, Prisma, and PostgreSQL.

6. Non-Functional Requirements & Security

- **B2B Payment Flow:** No payment processing on the platform; financial transactions are handled off-platform directly between PWC and the institutions.
- **Infrastructure:** Server space and architecture should be optimized for a fast, asset-light application handling text/form data without heavy media storage.

Security

- Enforce password reset on first login for all student accounts.
- Allow parent access only through signed, single-use, time-limited links.
- Lock pre-counselling forms after submission.
- Apply role-based access control:
 - Counsellors can access only assigned students.
 - Students can access only their own data and reports.
- Encrypt all personal and assessment data in transit and at rest.
- Maintain audit logs for report access, chart edits, and admin approvals.

Data Privacy & Retention

- Automatically purge data when a counselling cycle closes, with institution-level override support.
- Do not share data outside the platform except generated PDF reports.

Performance & Scalability

- Use a modular monolith for Phase 1, with clean module boundaries for Assessment, Scheduling, Career Library, and Reporting.
- Support 500 students per month, 30 sessions per day, and 10 concurrent sessions.
- Optimize for text/form-heavy usage, not media-heavy workloads.

- Index student ID, counsellor ID, and workflow status fields for faster reporting and dashboard queries.

7. Project Governance

- Regular project updates are shared by google meets & work progress tracker to be maintained.
- A staging environment maintained throughout, so PWC can review work-in-progress continuously rather than only at milestone sign-off
- Any feature request beyond the agreed scope as under section 'Phase 1 (MVP) Scope vs. Deferred Items' is logged and handled as a formal change request — not absorbed silently into the 30-day timeline.
- Up to 10% deviation from the agreed scope will be accommodated without triggering a change request; anything beyond that threshold is treated as a change request per the point above.
- Any delay with milestone completion will be communicated to the client team prior with the backup plans.
- Client to define & communicate the approval/signoff/review process in the starting of the project for the Neointeraction team to consider in the project plan.

Client detailed